

Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-7456
Project Name	ComicCraft – AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python-based load testing)
Type of Testing	Load Testing, Stress Testing and Spike Testing
Target Module / API	Backend endpoints: login, story script generation (Gemini text model), and comic panel image generation (Gemini image model)
Test Environment	[Local / Cloud / Hosted Server – specify]
Test Date	[dd-mm-yyyy]

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load test: users log in and open the home page / gallery	50	120	Avg response < 2 s, error rate < 1%
2	Load test: users submit story prompts to generate a comic script (Gemini text)	20	180	Script returned without failures; avg response within the agreed target for AI calls
3	Stress test: users generate panel images concurrently (Gemini image)	10 ramping to 50	300	App degrades gracefully; rate-limit/quota errors handled with clear messages
4	Spike test: sudden jump of users on login and gallery	10 to 100 in 10 s	60	System recovers after the spike; error rate < 1%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	[Fill after test]		
2	Response Time (Max)	< 5 seconds	[Fill after test]		
3	Throughput (Req/sec)	[e.g., ≥ 20 req/s]	[Fill after test]		
4	Error Rate	< 1%	[Fill after test]		
5	CPU Utilization	< 80%	[Fill after test]		

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
6	Memory Utilization	< 80%	[Fill after test]		

Note: Gemini text and especially image generation usually take longer than 2 s. Measure these calls separately (Scenarios 2 and 3) and state the target you applied in Remarks.

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about ComicCraft's performance, e.g., response time of script vs image generation]

Bottlenecks Identified:

[List observed bottlenecks, e.g., Gemini API latency, rate limits/quota, image size, database queries]

Optimization Steps Taken:

[Describe changes made, e.g., caching, async/queued image generation, retry with backoff, compressing images, loading indicators]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., Locust graphs and statistics tables):

